

Supplemental Figure Legends:

Supplemental Figure 1: ATAC-seq datasets; transcription factor motif enrichments and region similarities. Related to Figures 1 and 3.

(A-D): Log-transformed E-values of AME testing for motif enrichments of the indicated factors in each individual sequence set. Red horizontal line indicates significance ($p < 0.05$). Top specifically-enriched factors from each analysis are shown. (A) Enriched motifs when scanning 'general' (i.e. not mutually exclusive) sequence sets with our aggregated motif database. (B) Enriched motifs for 'general' sequences, using the set of differentially-expressed TFs. (C) Enriched motifs when scanning 'specific' (i.e. mutually-exclusive) sequence sets with our aggregated motif database. (D) Enriched motifs for 'specific' sequences, using the set of

differentially-expressed TFs. (E): Heatmap of Jaccard similarity index between late (E60s) samples, color scale indicates Z-transformed Jaccard index. Self-self comparisons are colored gray. (F) Similar to (E), for all early (E50s) and late (E60s) samples.

Supplemental Figure 2: Transcriptomic datasets. Related to Figures 2 and 3.

(A) PCA plot showing the distribution of all RNA-seq samples across the first two principal components. Tissues are colored as indicated, individual dots represent different individual samples. (B) PCA plot showing distribution of all RNA-seq samples across the first two principal components, using the top 1000 variable genes in the expression dataset. (C) Distance heatmap showing the distribution of all RNA-seq samples using distance between whole transcriptomes. Color scale indicates Euclidean distance between samples. (D) Volcano plot showing distribution of differentially-expressed genes identified in distal femur tissue (relative to all other bone-ends). The top five most strongly up/down-regulated genes are indicated. (E) Differential-expression boxplots showing the log₂-FC expression of individual tissues relative to the average cross-tissue expression of the indicated gene. Select genes with known roles in hindlimb development are shown. (F) GO BP enrichments for all genes upregulated in the distal femur. See Table S2 for details. (G) UMAP plot of single-cell expression datasets, integrated across three individual fetal long-bone samples (indicated with coloring). (H) UMAP plot of integrated dataset, with coloring indicating clustering of cells, with manually-annotated labels (see STAR Methods). (I) Estimated proportion of cell-types comprising each tissue-type from our bulk transcriptomic datasets, as calculated using our single-cell reference and MuSiC⁷⁵.

Supplemental Figure 3: Height associations and skeletal accessibility patterns. Related to

Figure 4. (A) Distribution of height-associated variants falling into bone-end-specific region sets (e.g. distal femur), or regions with epigenetic activity across multiple tissues. (Top/Bottom) Distribution for height-associated variants using a 1e-6/1e-8 cutoff. (B) LDSC enrichments for height associations across different E50's region sets. (C/D) Enrichments of E54 appendicular region sets for height variants defined at a p-value threshold of 1e-6/1e-8. (E/F) Enrichments of later E67 appendicular region sets for height variants defined at a p-value threshold of 1e-6/1e-8. (G) Enrichments of E67 appendicular region sets for COJO height variants. (H) LDSC enrichment scores for E67 sets. See also Table S3. (I) -log₁₀ p-values of variants binned into the indicated categories. (J) Distribution of variant -log₁₀ p-values based on the number of skeletal tissues in which accessibility was detected. Red dashed line indicates the linear regression model generated to fit these data. (K) The number of ATAC-seq overlaps (pooled across tissues) within a given independent height-associated locus (as defined in Yengo et al., 2023) were ranked and divided into deciles. The density of height-associated signals in independent loci was aggregated within deciles and plotted, yielding a significant linear-regression coefficient. (L) GO term enrichments for top 200 genes defined using our integrative ranking method (see STAR Methods). (M) GO term enrichments for top 200 genes defined using our integrative method, analyzed using different datasets; all epigenetic and height-association datasets (purple), all epigenetic and GWAS datasets, conditioned on interaction data (blue), height-association statistics only (green), and height-association statistics, conditioned on interaction data (red). For visualization, the latter two enrichment values are negated.

Supplemental Figure 4: Extending the resolution of GWAS using development epigenetics and transcriptomics. Related to Figure 4. (A) Diagrammatic illustration of approach to expand resolution within fine-mapped loci. (B, large pie chart) Proportion of fine-mapped loci in which the original fine-mapped variant (blue), linked variants (conditioned on skeletal accessibility) (red), both (purple), or neither (green) have PAINTOR posterior probabilities ≥ 0.99 . (Inset, small pie chart) Of those loci with posterior probabilities ≥ 0.99 , the proportion in which linked variants have greater predicted functionality than the original fine-mapped variant (see STAR Methods). (C) The proportion of genes initially assigned to fine-mapped loci ($n = 1190$) which were contacted by either the original fine-mapped variant ($p \geq 0.99$) (blue), PAINTOR significant linked variants (red), both (purple), or neither (green). (D) The proportion of unique genes contacted by original fine-mapped variants ($p \geq 0.99$) (blue), significant linked variants (red), or both (purple). (E) (Upper pie chart) The proportion of original fine-mapped variants looping to contact promoters (red), overlapping gene promoters directly (green), or not contacting any gene (blue). (Lower pie chart) Proportions for significant linked variants. (F) Example at *IGF2* for which epigenetics extends the resolution of the original fine-mapped locus. One of the original fine-mapped variants (far left) interacts with the *IGF2* promoter while an additional variant, identified through our PAINTOR analysis, contacts the same promoter. (G): Additional putative QTL activity at the *ECM1* locus. Top track: Promoter-contact interaction from Hi-C data. Rows 2-5 (descending): (2) COJO defined leads (3) PAINTOR alternate variants at the locus (4) GTEX QTLs which showed evidence of similar activity in our skeletal datasets (5) All significant GTEX QTLs associated with *ECM1*. Bottom: ATAC-seq peaks from our E54 datasets. In the *ECM1* locus, six eQTLs from our dataset had predicted regulatory effects (and were significant height-variants), with an additional four eQTLs overlapping chromatin interaction data. Of the *ECM1* eQTLs we detected, six are validated in the GTEX dataset, while the others represent new putative regulatory variants in chondrogenesis.

Supplemental Figure 5: Module definition, characterization, and network analyses. Related to Figure 5. (Left column) Gene Module Definition and CD1 Module characterization. (A) WGCNA plot showing hierarchical clustering of modules. (B) For genes in the CD1 module, the distribution of FPKM expression across different tissues – no significant differences were observed. (C) Distribution of average tissue-wise expression of genes within indicated modules. (D) Heatmap of module-module correlations, calculated as the percent of all pairwise gene comparisons for which significant negative correlations were observed. (E) PCA plot showing the distributions of modules based on the matrix of module-module correlation values. (F) Enrichments for height associations of indicated modules based on MAGMA (correlations between module eigenvectors and gene-wise association statistics via MAGMA). Negative values correspond to significant negative correlations between expression and association. Red lines indicate $FDR = 0.05$. (G) Genome-wide cumulative distribution of HESS block heritabilities. Red dots indicate blocks in the top 99% percentile of per-block assigned heritability. Red labels indicate CD1 module genes which are associated with these most-heritable blocks. (H) Genome-wide cumulative distribution of COJO window heritabilities (aggregated by block). Red dots indicate blocks in the top 99% percentile of per-block aggregated COJO heritability. Red labels indicate CD1 module genes which are associated with these most-heritable blocks. (I)

Distribution of per-gene assigned heritability based on HESS (expressed as a fraction of the total heritability captured by CD1 module genes). (J) Distribution of per-gene assigned heritability based on COJO windows (expressed as a fraction of the total heritability captured by CD1 module genes). (K) Distribution of FPKM values for genes falling within the indicated GO term (enriched in the CD1 module). (L) Distribution of per-gene heritabilities (expressed as a fraction of the total heritability captured by CD1 module genes), aggregated by GO term. (Middle Column) Motif Enrichments and SLDP Analysis. (M) Heatmap showing the enrichments for different transcription-factor motifs in putative regulatory regions around different module genes. (N) Per-motif enrichment plots. Red horizontal line indicates $-\log_{10}(0.05)$. (O) SLDP enrichments for variants hitting FOXP1 motifs around the indicated module sets. Y-axis is the calculated effect-size from SLDP results, adjusted p-value < 0.05 indicated with '*'. (P) PCA plot of GO terms enriched for nearby FOXP1-altering SNPs. Coloring indicates terms grouped by semantic clustering. Size of points is scaled by calculated effect-size from SLDP results. See Table S5. (Right Column) Effect-size (relative and absolute) distributions for module-network analysis. Clustering these enriched terms by semantic similarity, we saw several enriched GO categories relating to development of various tissues (including ossification terms), as well as general cellular processes (e.g., 'regulation of growth'). (Q) Distributions of aggregated locus-wise height effect-size (beta) for the indicated module in comparison to a randomized module set, or OMIM skeletal genes. See Table S5. (R) Absolute effect-size (beta) distributions for module-network analysis. Distributions of aggregated locus-wise height absolute effect-size (beta) for the indicated module in comparison to a randomized module set, or OMIM skeletal genes.

Supplemental Figure 6: Additional information for application of framework to genetic associations of Type 2 Diabetes. Related to Figure 6. (A) Enrichments for Type 2 Diabetes associations of indicated modules based on MAGMA. Horizontal dashed lines indicate $FDR = 0.05$. (B) Comparisons of set-averaged values for modules relative to size-matched background distributions (see STAR Methods, Table S6). Asterix next to module name indicates significant enrichment. (Left) Distribution of per-locus averaged $-\log_{10}(p\text{-value})$ for a given module, compared to the distribution obtained from a randomly generated, size-matched 'module'. Top P-value shown is the rate at which a randomized module has a significantly greater distribution than the indicated module via t-test. Bottom p-value is the frequency of random modules with an average value greater than the target set. For depletions, bottom p-value is calculated as $1 - (\text{frequency})$. (Middle) Same as Left, using per-locus averaged signed effect size (beta coefficient). (Right) Same as Left/Middle, using per-locus absolute effect size (beta coefficient).











